

WASP-3b

R-1121

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-3_141013 · created 2026-07-30T21:51:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456943.6561 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.071 +/- 0.019
Transit depth (Rp/Rs)^2	0.51 +/- 0.27 [%]
Orbital Inclination (inc)	85.9 +/- 2.62
Airmass coefficient 1 (a1)	1.2938 +/- 0.009
Airmass coefficient 2 (a2)	-0.1435 +/- 0.0031
Scatter in the residuals of the lightcurve fit is	0.69 %
Best Comparison Star	None
Optimal Aperture	7.54
Optimal Annulus	9.42
Transit Duration (day)	0.116 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.071 ± 0.019 vs 0.10538 (deviation 1.81σ)
Duration fit vs published	0.116 d vs 0.1167 d (deviation 0.06σ)
Mid-time O–C	4.1 min
Depth SNR	3.7
Residual scatter	0.69 %

Light curve

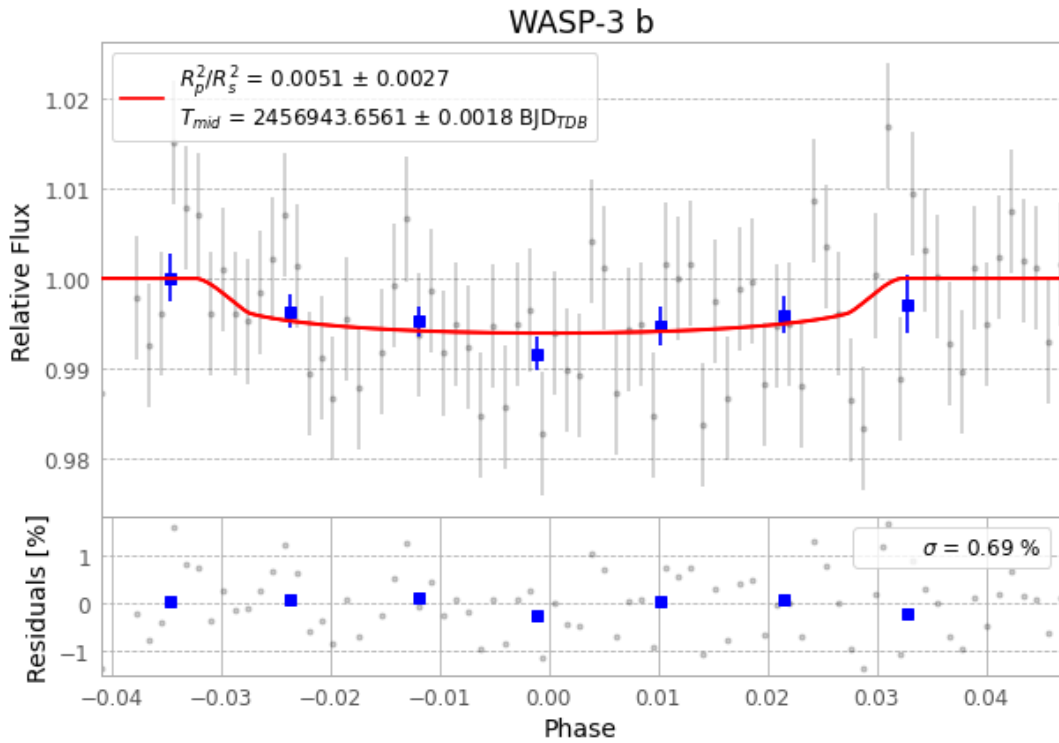


Figure 1 — FinalLightCurve_WASP-3 b_2014-10-12.png (SHA-256 bc54678347c220d4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [462, 180], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6b5ed18868c8335a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-3141013015443.FITS → WASP-3141013054812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-141013.log (63a13fa937e4...), FinalLightCurve_WASP-3 b_2014-10-12.png (bc54678347c2...), FinalLightCurve_WASP-3 b_2014-10-12.csv (9c08151289c9...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 21:51Z.